SOFTWARE ENGINEERING LAB 6

CODE AND BRANCH COVERAGE ANALYSIS

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1. Objective

The objective of this experiment is to understand and analyse code coverage and branch coverage using Python's Coverage.py tool. The experiment involves writing additional test cases, identifying untested code paths, and improving test coverage to achieve at least 90% coverage.

2. Introduction

Code coverage is a software testing metric that measures how much of the source code is executed during testing.

The two main types of coverage analysed in this experiment are:

- **Code Coverage:** Measures the percentage of executable statements that are executed during testing.
- **Branch Coverage:** Measures whether both possible outcomes of decision-making statements, such as if-else conditions, are executed.

Branch coverage helps identify untested logical paths that may not be detected through statement coverage alone.

3. Tools and Technologies Used

- Python 3.13.7
- Pytest 9.2.0
- Coverage.py 7.17.0
- Windows PowerShell

4. Application Description

The assigned application is a Movie Ticket Booking system implemented in Python.

The application contains three main functions:

1. ticket_price(seat, age, show_hour, is_weekend)

Calculates movie ticket prices based on seat category, customer age, show timing, and whether the show is scheduled on a weekend.

2. group_discount(tickets, is_student_group)

Calculates discounts based on the number of tickets booked and whether the booking belongs to a student group.

3. snack_combo(size, has_coupon)

Calculates snack combo prices based on the selected size and coupon availability.

5. Initial Coverage Analysis

Initially, the application was tested using the two test cases provided in the assigned repository.

The initial test execution was successful, but several statements and branches remained untested.

Initial Code Coverage:

- Application statement coverage: 50%
- Missing statements: 19 out of 38

Initial Branch Coverage:

- Application branch coverage: 45%
- Total branches: 28
- Partially covered branches: 11

These results indicated that the initial test cases were insufficient to cover all application logic.

6. Code Coverage Analysis

An additional test file named `CodeCoverageTest_454_movie_tickets.py` was created.

A total of 15 test cases were implemented to test the three application functions.

The test cases covered:

- Invalid seat categories.
- Invalid show timings.
- Morning and evening show pricing.
- Weekend pricing.
- Child ticket pricing and age-based discounts.

- Different group sizes and student discounts.
- Different snack sizes and coupon conditions.

The tests were executed using pytest and Coverage.py.

Final Code Coverage Result:

- Total statements: 38
- Executed statements: 38
- Missing statements: 0
- Code coverage achieved: 100%
- Tests passed: 15

7. Branch Coverage Analysis

A separate test file named `BranchCoverageTest_454_movie_tickets.py` was created to test different decision-making paths in the application.

A total of 23 test cases were implemented.

The test cases covered:

- Valid and invalid seat categories.
- Valid and invalid show timings.
- Morning, afternoon, and evening show conditions.
- Weekend and weekday conditions.
- Different age categories.
- All group discount conditions.
- Student and non-student group conditions.
- Different snack sizes.
- Coupon and non-coupon conditions.

Branch coverage was measured by executing pytest with the `--branch` option.

Final Branch Coverage Result:

- Total statements: 38
- Missing statements: 0
- Total branches: 28
- Partially covered branches: 0
- Branch coverage achieved: 100%
- Tests passed: 23

8. Comparison of Initial and Final Results

Metric	Initial Coverage	Final Coverage
Code Coverage	50%	100%
Branch Coverage	45%	100%

The additional test cases successfully covered all executable statements and decision branches in the application.

9. Conclusion

The experiment demonstrated the importance of code coverage and branch coverage in software testing.

Initially, the application had incomplete test coverage because only two test cases were provided. By developing additional test cases using pytest and analysing the results using Coverage.py, both statement coverage and branch coverage were improved to 100%.

The experiment helped in understanding how systematic testing can identify untested code paths, improve test completeness, and increase confidence in application behaviour.

10. Deliverables

1. Application source file: `movie_tickets.py`
2. Code coverage test file: `CodeCoverageTest_454_movie_tickets.py`
3. Branch coverage test file: `BranchCoverageTest_454_movie_tickets.py`
4. Screenshots containing initial and final coverage results.
5. GitHub commits documenting the code coverage and branch coverage test cases.

SCREENSHOTS:

```
None
01_Original_Application.png
```

```
PS C:\Users\Varsh\Desktop\PES1UG24AM454_SE_LAB\PES1UG24AM454_Lab6> cd "$HOME\Desktop\L6_L7_15_movie_tickets\Lab 6"
PS C:\Users\Varsh\Desktop\L6_L7_15_movie_tickets\Lab 6> py movie_tickets.py
PS C:\Users\Varsh\Desktop\L6_L7_15_movie_tickets\Lab 6> py -c "from movie_tickets import *; print('Regular ticket:', tic
ket_price('regular', 25, 15, False)); print('Premium ticket:', ticket_price('premium', 25, 20, True)); print('Group disc
ount:', group_discount(10, True)); print('Snack combo:', snack_combo('small', True))"
Regular ticket: 150
Premium ticket: 320
Group discount: 20
Snack combo: 70
```

None

02_Initial_Code_Coverage.png

```
PS C:\Users\Varsh\Desktop\PES1UG24AM454_SE_LAB\PES1UG24AM454_Lab6> py -m coverage erase
PS C:\Users\Varsh\Desktop\PES1UG24AM454_SE_LAB\PES1UG24AM454_Lab6> py -m coverage run -m pytest test_movie_tickets.py
============================================= test session starts =============================================
platform win32 -- Python 3.13.7, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\Varsh\Desktop\PES1UG24AM454_SE_LAB\PES1UG24AM454_Lab6
plugins: anyio-4.10.0, cov-7.1.0
collected 2 items

test_movie_tickets.py ..                                                                              [100%]

============================================== 2 passed in 0.06s ==============================================
PS C:\Users\Varsh\Desktop\PES1UG24AM454_SE_LAB\PES1UG24AM454_Lab6> py -m coverage report -m
Name                   Stmts   Miss  Cover   Missing
----------------------------------------------------
movie_tickets.py          38     19    50%   6, 8, 11, 13, 15, 17, 19, 25, 27, 29, 33, 38-46
test_movie_tickets.py      6      0   100%
----------------------------------------------------
TOTAL                     44     19    57%
PS C:\Users\Varsh\Desktop\PES1UG24AM454_SE_LAB\PES1UG24AM454_Lab6>
```

None

03_Initial_Branch_Coverage.png

```
PS C:\Users\Varsh\Desktop\PES1UG24AM454_SE_LAB\PES1UG24AM454_Lab6> py -m coverage erase
PS C:\Users\Varsh\Desktop\PES1UG24AM454_SE_LAB\PES1UG24AM454_Lab6> py -m coverage run --branch -m pytest test_movie_tick
ets.py
============================================= test session starts =============================================
platform win32 -- Python 3.13.7, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\Varsh\Desktop\PES1UG24AM454_SE_LAB\PES1UG24AM454_Lab6
plugins: anyio-4.10.0, cov-7.1.0
collected 2 items

test_movie_tickets.py ..                                                                              [100%]

============================================== 2 passed in 0.05s ==============================================
PS C:\Users\Varsh\Desktop\PES1UG24AM454_SE_LAB\PES1UG24AM454_Lab6> py -m coverage report -m
Name                   Stmts   Miss Branch BrPart  Cover   Missing
------------------------------------------------------------------
movie_tickets.py          38     19     28     11    45%   6, 8, 11, 13, 15, 17, 19, 25, 27, 29, 33, 38-46
test_movie_tickets.py      6      0      0      0   100%
------------------------------------------------------------------
TOTAL                     44     19     28     11    50%
PS C:\Users\Varsh\Desktop\PES1UG24AM454_SE_LAB\PES1UG24AM454_Lab6>
```

None

04_Final_Code_Coverage.png

```
PS C:\Users\Varsh\Desktop\PES1UG24AM454_SE_LAB\PES1UG24AM454_Lab6> notepad CodeCoverageTest_454_movie_tickets.py
PS C:\Users\Varsh\Desktop\PES1UG24AM454_SE_LAB\PES1UG24AM454_Lab6> py -m coverage erase
PS C:\Users\Varsh\Desktop\PES1UG24AM454_SE_LAB\PES1UG24AM454_Lab6> py -m coverage run --include="movie_tickets.py" -m py
test CodeCoverageTest_454_movie_tickets.py
============================================= test session starts =============================================
platform win32 -- Python 3.13.7, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\Varsh\Desktop\PES1UG24AM454_SE_LAB\PES1UG24AM454_Lab6
plugins: anyio-4.10.0, cov-7.1.0
collected 15 items

CodeCoverageTest_454_movie_tickets.py ...............                                                    [100%]

============================================== 15 passed in 0.44s ==============================================
PS C:\Users\Varsh\Desktop\PES1UG24AM454_SE_LAB\PES1UG24AM454_Lab6> py -m coverage report -m --include="movie_tickets.py"

Name               Stmts   Miss  Cover   Missing
------------------------------------------------
movie_tickets.py      38      0   100%
------------------------------------------------
TOTAL                 38      0   100%
PS C:\Users\Varsh\Desktop\PES1UG24AM454_SE_LAB\PES1UG24AM454_Lab6>
```

None

05_Final_Branch_Coverage.png

```
PS C:\Users\Varsh\Desktop\PES1UG24AM454_SE_LAB\PES1UG24AM454_Lab6> notepad BranchCoverageTest_454_movie_tickets.py
PS C:\Users\Varsh\Desktop\PES1UG24AM454_SE_LAB\PES1UG24AM454_Lab6> py -m coverage erase
PS C:\Users\Varsh\Desktop\PES1UG24AM454_SE_LAB\PES1UG24AM454_Lab6> py -m coverage run --branch --include="movie_tickets.
py" -m pytest BranchCoverageTest_454_movie_tickets.py
============================================= test session starts =============================================
platform win32 -- Python 3.13.7, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\Varsh\Desktop\PES1UG24AM454_SE_LAB\PES1UG24AM454_Lab6
plugins: anyio-4.10.0, cov-7.1.0
collected 23 items

BranchCoverageTest_454_movie_tickets.py .......................                                          [100%]

============================================== 23 passed in 0.10s ==============================================
PS C:\Users\Varsh\Desktop\PES1UG24AM454_SE_LAB\PES1UG24AM454_Lab6> py -m coverage report -m --include="movie_tickets.py"

Name               Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------
movie_tickets.py      38      0     28      0   100%
-----------------------------------------------------------------
TOTAL                 38      0     28      0   100%
PS C:\Users\Varsh\Desktop\PES1UG24AM454_SE_LAB\PES1UG24AM454_Lab6>
```